Detecting Money Laundering in Transaction Networks

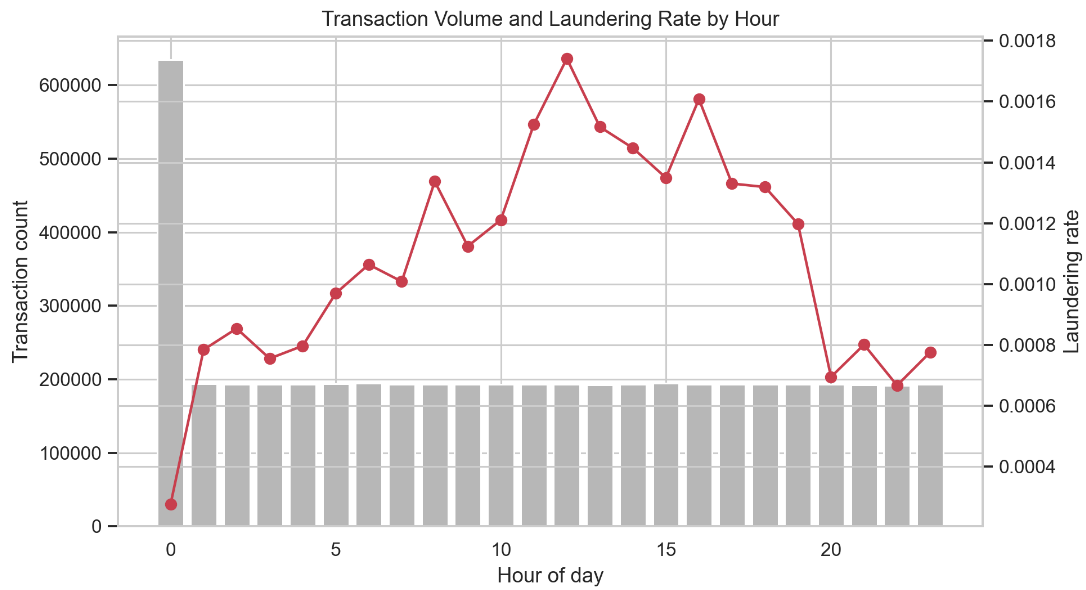
- From tabular ML to graph-aware AML features.
- Dataset: IBM synthetic AML transactions.
- Goal: build an interpretable rare-event detection pipeline.

Why AML Detection Is Hard

- Only 0.1019% of all transactions are laundering.
- A naive all-legitimate classifier has high accuracy but zero usefulness.
- We prioritize recall, F1, ROC-AUC, and PR-AUC.

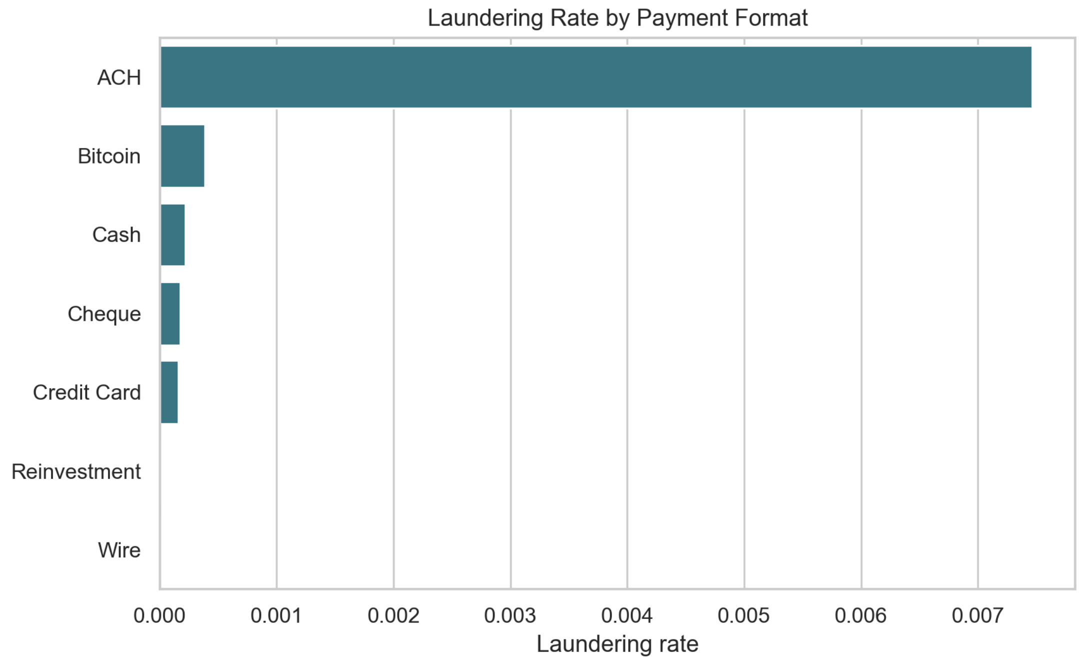
Dataset and Target

- Full data: 5,078,345 transactions.
- Laundering labels: 5,177.
- Target: Is Laundering, a binary transaction-level label.



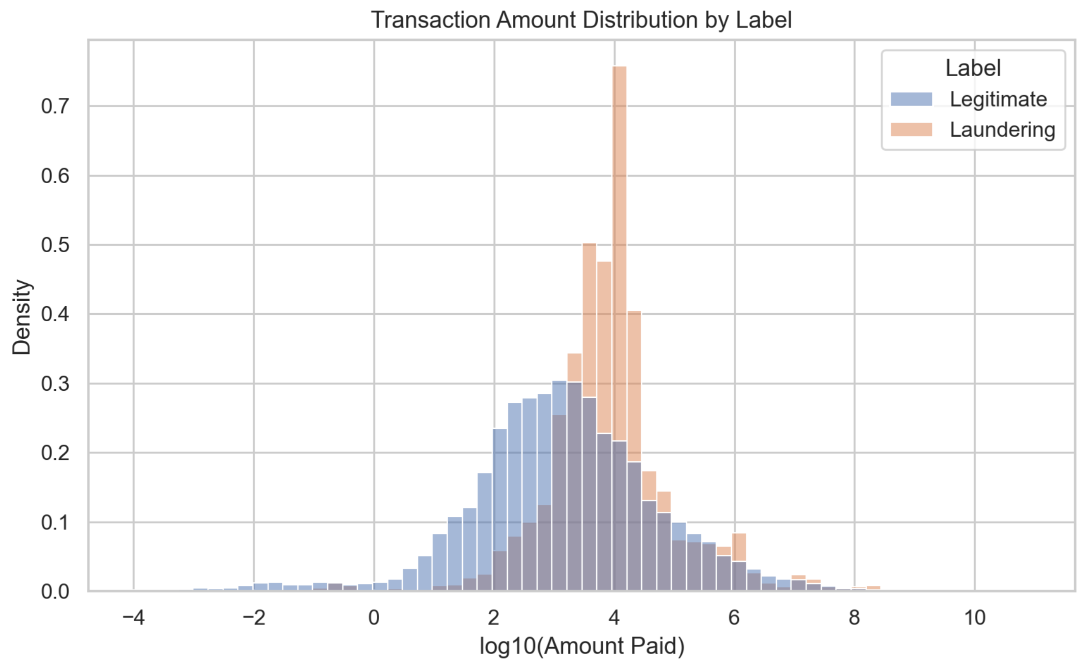
Feature Engineering

- Temporal features: hour, weekday, weekend.
- Amount and currency features: log amount, amount difference, same currency.
- Graph features: account degrees, unique counterparties, totals, PageRank.



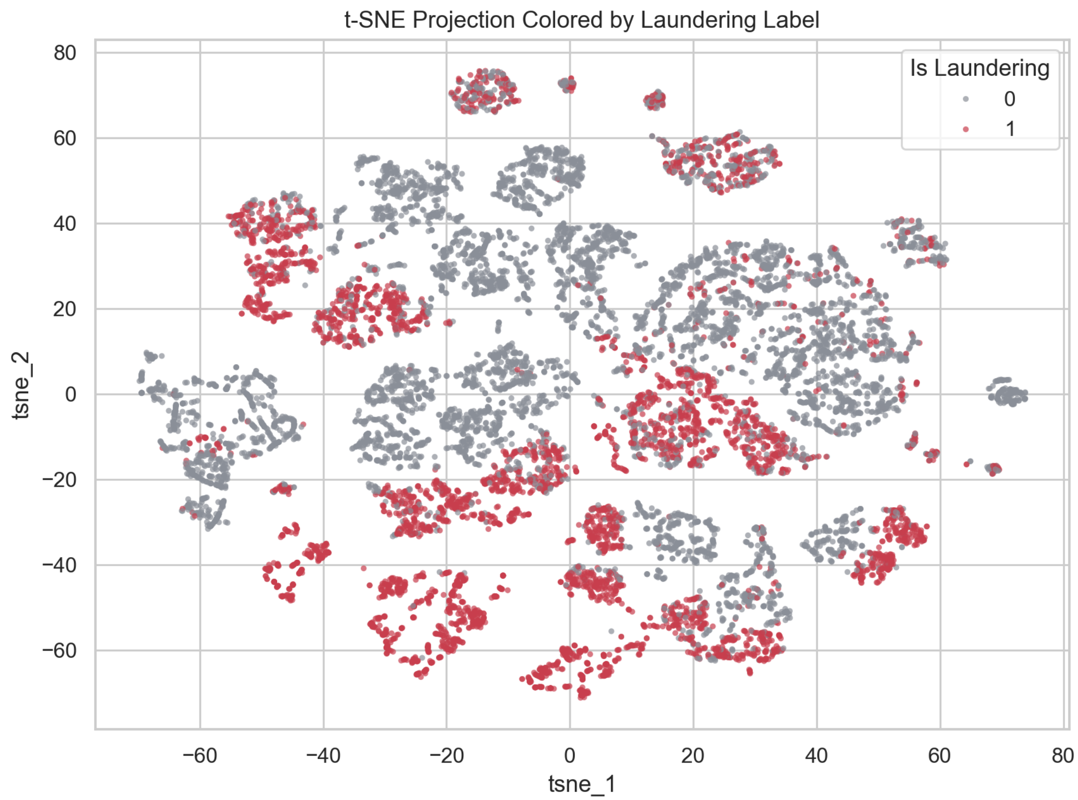
Exploratory Finding

- ACH transactions have the highest laundering rate.
- Amount distribution becomes more interpretable after log transformation.
- EDA guides the feature design and model interpretation.



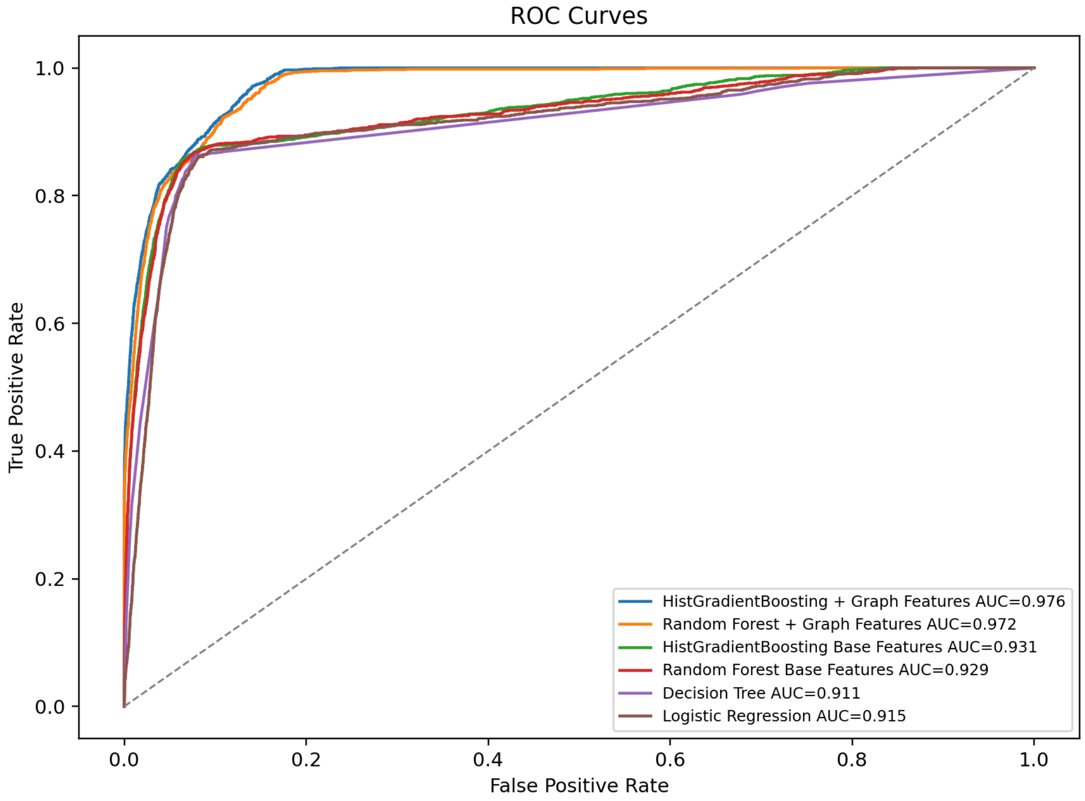
t-SNE and Clustering

- t-SNE reveals local structure in the enriched sample.
- MiniBatch K-Means and DBSCAN identify high-risk regions.
- Clusters are used for pattern discovery, not as ground truth.



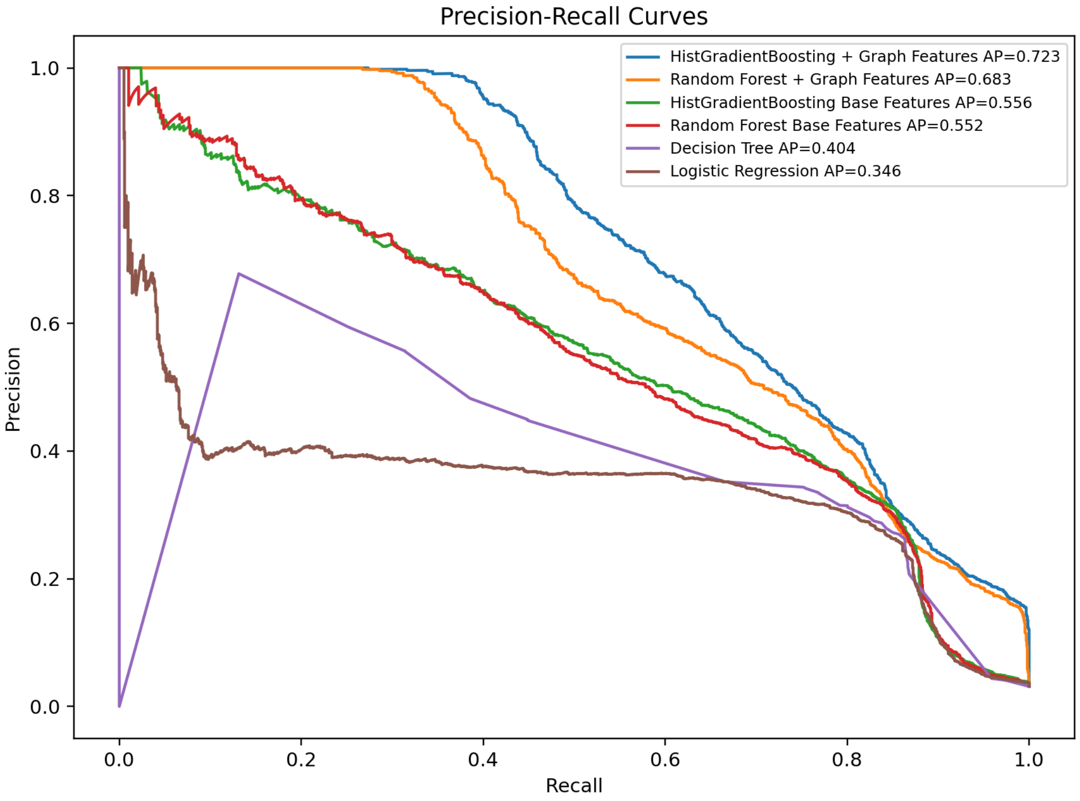
Model Comparison Setup

- Baselines: Logistic Regression and Decision Tree.
- Strong models: Random Forest and HistGradientBoosting.
- Ablation: base features vs base + graph-aware features.



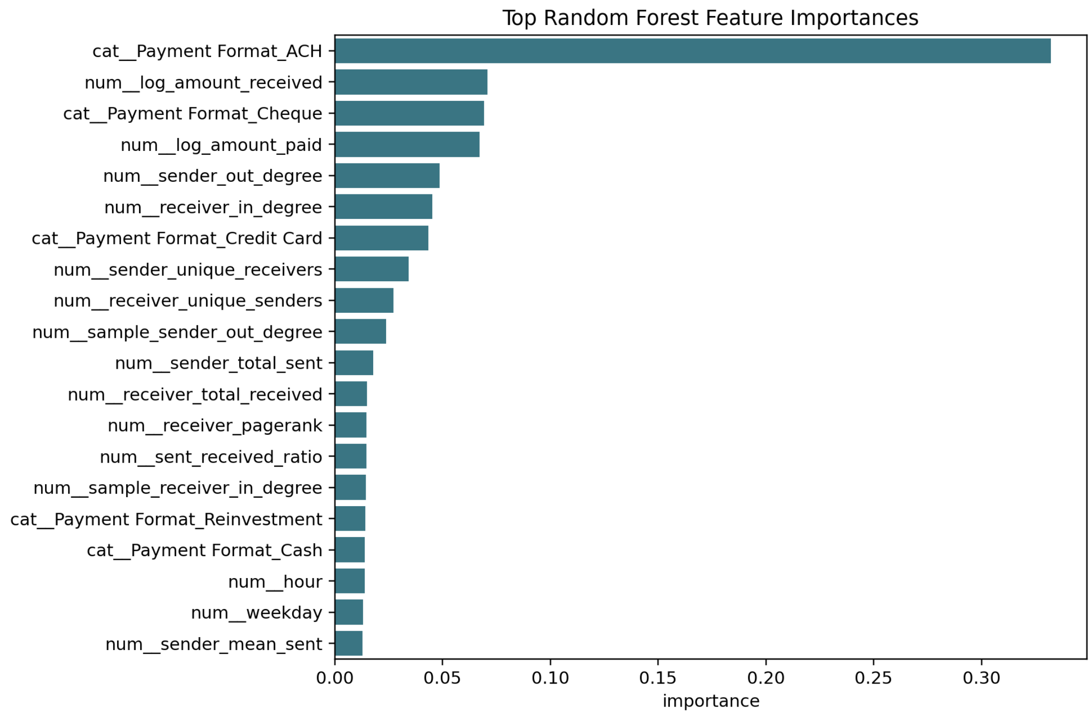
Main Result

- Best model: HistGradientBoosting + Graph Features.
- F1 = 0.642.
- ROC-AUC = 0.976; PR-AUC = 0.723.



Graph Feature Impact

- HGB base F1: 0.548; HGB + graph F1: 0.642.
- HGB base PR-AUC: 0.556; HGB + graph PR-AUC: 0.723.
- Important features include payment format, log amount, sender degree, and receiver degree.



Conclusion

- Accuracy is insufficient for AML because the positive class is extremely rare.

- Graph-aware features improve minority-class detection.

- Future work: time-based validation, subgraph mining, and temporal GNNs.